AMERICAN ECONOMIC DATA REPOSITORY (AEDR)

Detailed Labor Database

Technical Reference & Data Dictionary

QCEW + NAICS + OEWS + SOC

Repository domain: 02_Labor_Income

Coverage: QCEW 1990–2025 | OEWS 2011–2025

Prepared August 2026 | Revised to include finalized OEWS analytical and visualization views

Purpose

This document explains how the completed AEDR Detailed Labor Database is organized, what each source contributes, how NAICS and SOC supply hierarchical meaning, and how to interpret the principal analytical fields and views. This revision adds the finalized OEWS analytical architecture, including the occupation-classification layer and the visualization-ready OEWS view.

1. Database Purpose and Relational Structure

The Detailed Labor Database combines two complementary U.S. Bureau of Labor Statistics (BLS) programs. QCEW describes employment, establishments, wages, ownership, geography, and industry structure. OEWS describes employment and wage distributions by occupation, geography, and industry. NAICS supplies the industry hierarchy used to interpret QCEW and industry dimensions in OEWS; SOC supplies the occupation hierarchy used to interpret OEWS.

Component	Role in the database	Primary question answered
QCEW	Core industry/employer-side labor dataset	What is happening to employment, establishments, and wages within industries and places?
NAICS	Industry classification/reference system	What industry does a code represent, and where does it sit in the industry hierarchy?
OEWS	Core occupation-side employment and wage dataset	What is happening to particular occupations, their employment, and their wage distributions?
SOC	Occupation classification/reference system	What occupation does a code represent, and where does it sit in the occupation hierarchy?

Conceptual relationship:

- QCEW → Geography + Ownership + Industry code → NAICS hierarchy
- OEWS → Geography + Industry/NAICS + Occupation code → SOC hierarchy
- QCEW and OEWS complement one another; they are not interchangeable measures.

2. QCEW: Industry, Establishments, Employment, and Wages

The Quarterly Census of Employment and Wages (QCEW) is primarily based on state unemployment-insurance administrative records. BLS reports that QCEW covers more than 95 percent of U.S. jobs and publishes employment and wage information by detailed industry at national, state, county, and other supported geographic levels. In AEDR, the annual single-file series from 1990 through 2025 was validated, normalized, and consolidated into a 100,006,739-row canonical table.

A QCEW row is not simply an "industry-year." Its identifying grain is the combination of year, annual-quarter marker, area, ownership, industry, aggregation level, and establishment-size class. The AEDR normalized master table preserves this structure.

Key dimension	Meaning
area_fips	BLS QCEW geographic area identifier; may be numeric-looking or alphanumeric.
own_code	Ownership classification, such as private or levels of government.
industry_code	BLS/NAICS industry or higher-level BLS industry code.
agglvl_code	Defines the geographic and industry resolution represented by the row.
size_code	Establishment-size classification.
year / qtr	Reference year and annual marker (qtr = A in annual files).

3. NAICS: How QCEW Drills Down by Industry

The North American Industry Classification System (NAICS) gives QCEW industry codes their hierarchical meaning. AEDR does not interpret every historical observation with the current classification. Instead, it assigns the historically appropriate NAICS vintage and exposes parent classifications through a unified hierarchy reference.

Observation years	NAICS vintage used in AEDR
1990–2006	2002
2007–2010	2007
2011–2016	2012
2017–2021	2017
2022–2025	2022

Industry drill-down used by the database: All Industries → BLS Domain → BLS Supersector → NAICS Sector (2-digit) → Subsector (3-digit) → Industry Group (4-digit) → NAICS Industry (5-digit) → National Industry (6-digit)

Comparisons across classification boundaries should account for revisions to NAICS.

4. QCEW AEDR Tables and Views

Object	Type / purpose
qcew_annual	Canonical normalized master table. Consolidates 36 annual source tables (1990–2025) and preserves all 38 source-data fields.

qcew_annual_readable	Human-readable reference view adding official BLS titles for area, ownership, industry, aggregation level, and size.
ref_naics_codes	Reference layer supporting industry-code interpretation.
ref_naics_hierarchy	Unified historical NAICS hierarchy across 2002, 2007, 2012, 2017, and 2022 vintages.
qcew_annual_enriched	Enriched analytical layer joining normalized observations to historical classification metadata.
qcew_annual_final	Primary human-readable analytical view for downstream research and visualization.

Engineering note: numeric QCEW area identifiers were restored to five-character text values when needed (for example, 01000), while valid alphanumeric area codes were preserved. Final reference validation found zero unmatched area, ownership, industry, aggregation-level, or size titles.

5. QCEW Core Data Dictionary

Field	Storage	Origin	Interpretation
area_fips	TEXT	Source	5-character QCEW area identifier; geographic key.
own_code	INTEGER/TEXT	Source	Ownership code.
industry_code	TEXT	Source	Industry/BLS classification code; interpreted with appropriate historical NAICS vintage.
agglvl_code	INTEGER/TEXT	Source	Aggregation-level code defining geographic and industry resolution.
size_code	INTEGER/TEXT	Source	Establishment-size class code.
year	INTEGER	Source	Reference year.
qtr	TEXT	Source	Annual file marker; normally A.
disclosure_code	TEXT	Source	BLS disclosure/suppression indicator.
annual_avg_estabs	INTEGER	Source	Annual average of quarterly establishment counts.
annual_avg_emplvl	INTEGER	Source	Annual average of monthly employment levels; not unique persons employed during the year.
total_annual_wages	INTEGER	Source	Sum of the four quarterly total-wage levels.
taxable_annual_wages	INTEGER	Source	Sum of quarterly taxable wages.
annual_contributions	INTEGER	Source	Sum of quarterly unemployment-insurance contribution totals.
annual_avg_wkly_wage	INTEGER	Source	BLS average weekly wage based on annual wages and monthly employment levels.
avg_annual_pay	INTEGER	Source	BLS average annual pay based on employment and wage levels.
lq_disclosure_code	TEXT	Source	Disclosure indicator for location-quotient measures.
lq_annual_avg_estabs	REAL	Source	Location quotient for annual-average establishments relative to the U.S.
lq_annual_avg_emplvl	REAL	Source	Location quotient for annual-average employment relative to the U.S.
lq_total_annual_wages	REAL	Source	Location quotient for total annual wages relative to the U.S.
lq_taxable_annual_wages	REAL	Source	Location quotient for taxable annual wages relative to the U.S.
lq_annual_contributions	REAL	Source	Location quotient for annual contributions relative to the U.S.
lq_annual_avg_wkly_wage	REAL	Source	Location quotient for average weekly wage relative to the U.S.
lq_avg_annual_pay	REAL	Source	Location quotient for average annual pay relative to the U.S.
oty_disclosure_code	TEXT	Source	Disclosure indicator for over-the-year measures.
oty_annual_avg_estabs_chg	INTEGER/REAL	Source	Over-the-year change in annual-average establishments.
oty_annual_avg_estabs_pct_chg	REAL	Source	Over-the-year percent change in annual-average establishments.
oty_annual_avg_emplvl_chg	INTEGER/REAL	Source	Over-the-year change in annual-average employment.
oty_annual_avg_emplvl_pct_chg	REAL	Source	Over-the-year percent change in annual-average employment.
oty_total_annual_wages_chg	INTEGER/REAL	Source	Over-the-year change in total annual wages.
oty_total_annual_wages_pct_chg	REAL	Source	Over-the-year percent change in total annual wages.
oty_taxable_annual_wages_chg	INTEGER/REAL	Source	Over-the-year change in taxable annual wages.
oty_taxable_annual_wages_pct_chg	REAL	Source	Over-the-year percent change in taxable annual wages.
oty_annual_contributions_chg	INTEGER/REAL	Source	Over-the-year change in annual contributions.
oty_annual_contributions_pct_chg	REAL	Source	Over-the-year percent change in annual contributions.
oty_annual_avg_wkly_wage_chg	INTEGER/REAL	Source	Over-the-year change in average weekly wage.
oty_annual_avg_wkly_wage_pct_chg	REAL	Source	Over-the-year percent change in average weekly wage.
oty_avg_annual_pay_chg	INTEGER/REAL	Source	Over-the-year change in average annual pay.
oty_avg_annual_pay_pct_chg	REAL	Source	Over-the-year percent change in average annual pay.

6. QCEW Enrichment Fields

The readable/final views add descriptive fields rather than replacing original coded dimensions. Depending on the view, these include area_title, own_title, industry_title, agglvl_title, size_title, assigned NAICS vintage, geographic-level descriptions, industry-level descriptions, and NAICS parent hierarchy fields (domain, supersector, sector, subsector, industry group, NAICS industry, and national industry). These are reference/derived fields layered above preserved QCEW measures.

7. OEWS: Occupations, Employment, and Wage Distributions

The Occupational Employment and Wage Statistics (OEWS) program produces annual employment and wage estimates by occupation. It supports national, state, metropolitan/nonmetropolitan, and industry-specific analysis. In AEDR, the consolidated annual series covers 2011–2025 and contains 6,196,391 observations.

OEWS complements QCEW by changing the analytical lens. QCEW is strongest for industry/employer structure; OEWS is strongest for occupation/job structure. An OEWS observation is identified by year plus its geography, industry/ownership context, occupation code, and reporting-group dimensions.

QCEW lens	OEWS lens

Industry and employer structure	Occupation and job structure
Establishments, employment, wages	Occupational employment and wage distribution
NAICS is the principal industry hierarchy	SOC is the principal occupation hierarchy; NAICS also describes industry-specific estimates

8. SOC: How OEWS Drills Down by Occupation

The Standard Occupational Classification (SOC) system supplies the formal occupational hierarchy. AEDR retains both the 2010 and 2018 SOC reference systems and the official 2010→2018 crosswalk so historical OEWS codes are not interpreted as though the classification never changed.

SOC hierarchy: Major Group → Minor Group → Broad Group → Detailed Occupation

Reference	Validated AEDR contents
2010 SOC structure	1,421 hierarchy rows: 23 major, 97 minor, 461 broad, 840 detailed.
2010 SOC definitions	840 detailed occupations; exact correspondence with detailed structure codes.
2018 SOC structure/definitions	1,447 hierarchy codes: 23 major, 98 minor, 459 broad, 867 detailed.
2010→2018 crosswalk	900 relationships; 840 unique 2010 endpoints and 867 unique 2018 endpoints; no invalid endpoints or duplicate code pairs.

OEWS contains legitimate BLS reporting codes that do not always correspond one-for-one to a formal detailed SOC occupation. Examples include All Occupations (00-0000), aggregate/group codes, historical modified codes, and transition/combined categories. AEDR therefore preserves OEWS occ_code exactly as published and uses SOC as a reference/classification system rather than as a restrictive validity constraint.

9. OEWS AEDR Tables and Views

Object	Type / purpose
oews_annual	Canonical normalized OEWS master table containing the consolidated 2011–2025 source series. It preserves the 52 normalized source fields and the original published OEWS codes, titles, measures, flags, and reporting dimensions.
ref_oews_occupation_classification	Occupation-classification reference layer used to resolve formal SOC occupations and legitimate OEWS-specific, aggregate, hybrid, and historical reporting categories without deleting valid source observations.
oews_annual_final	Primary enriched analytical OEWS view. Preserves all 52 oews_annual fields and adds occupation classification, SOC version, hierarchy, and OEWS-specific reference metadata. The view contains 6,196,391 rows, exactly matching the canonical source table.
oews_visualization_final	Visualization/BI projection of oews_annual_final. Contains 79 columns and presents a simplified occupation hierarchy alongside parallel technical SOC/OEWS classification fields. It performs no filtering, aggregation, grouping, deduplication, or additional joins.

Design rule: oews_annual_final does not directly join the historical 2010→2018 SOC crosswalk because one-to-many crosswalk relationships could duplicate observations. The crosswalk remains a reference tool for historical interpretation rather than a row-generating analytical join.

For most occupation-level research, use oews_annual_final when full analytical metadata is required and oews_visualization_final when building Tableau or other BI visualizations. The visualization view is intentionally a column-selection layer, so its row-level meaning remains identical to oews_annual_final.

10. OEWS Core Data Dictionary

Field	Storage	Interpretation
year	INTEGER	Reference year.
area	TEXT	OEWS geographic area code.
area_title	TEXT	Human-readable geographic title.
area_type	INTEGER	OEWS area-type classification.
prim_state	TEXT	Primary state abbreviation/code where applicable.
naics	TEXT	Industry code or OEWS industry context.
naics_title	TEXT	Industry title.
i_group	TEXT	Industry grouping/reporting level.
own_code	TEXT	Ownership code/context.
occ_code	TEXT	Occupation/reporting code; preserved exactly as published.
occ_title	TEXT	Occupation/reporting title.
o_group	TEXT	Occupational reporting level/group; includes total/major/minor/broad/detailed and legacy patterns.
tot_emp	INTEGER	Estimated occupational employment.
tot_emp_flag	TEXT	Flag associated with total employment.
emp_prse	REAL	Percent relative standard error for employment estimate.
emp_prse_flag	TEXT	Flag associated with employment PRSE.
jobs_1000	REAL	Employment per 1,000 jobs where applicable.
jobs_1000_flag	TEXT	Flag for jobs-per-1,000 measure.
loc_quotient	REAL	Occupational location quotient.

loc_quotient_flag	TEXT	Flag for location quotient.
pct_total	REAL	Percent of total employment where applicable.
pct_total_flag	TEXT	Flag for percent-of-total measure.
pct_rpt	REAL	Percent reported measure where applicable.
pct_rpt_flag	TEXT	Flag for percent-reported measure.
h_mean	REAL	Mean hourly wage.
h_mean_flag	TEXT	Flag for mean hourly wage.
a_mean	INTEGER	Mean annual wage.
a_mean_flag	TEXT	Flag for mean annual wage.
mean_prse	REAL	Percent relative standard error for mean wage.
mean_prse_flag	TEXT	Flag for mean-wage PRSE.
h_pct10	REAL	10th-percentile hourly wage.
h_pct10_flag	TEXT	Flag for 10th-percentile hourly wage.
h_pct25	REAL	25th-percentile hourly wage.
h_pct25_flag	TEXT	Flag for 25th-percentile hourly wage.
h_median	REAL	Median hourly wage.
h_median_flag	TEXT	Flag for median hourly wage.
h_pct75	REAL	75th-percentile hourly wage.
h_pct75_flag	TEXT	Flag for 75th-percentile hourly wage.
h_pct90	REAL	90th-percentile hourly wage.
h_pct90_flag	TEXT	Flag for 90th-percentile hourly wage.
a_pct10	INTEGER	10th-percentile annual wage.
a_pct10_flag	TEXT	Flag for 10th-percentile annual wage.
a_pct25	INTEGER	25th-percentile annual wage.
a_pct25_flag	TEXT	Flag for 25th-percentile annual wage.
a_median	INTEGER	Median annual wage.
a_median_flag	TEXT	Flag for median annual wage.
a_pct75	INTEGER	75th-percentile annual wage.
a_pct75_flag	TEXT	Flag for 75th-percentile annual wage.
a_pct90	INTEGER	90th-percentile annual wage.
a_pct90_flag	TEXT	Flag for 90th-percentile annual wage.
annual	TEXT	OEWS annual/hourly reporting indicator retained from source schema.
hourly	TEXT	OEWS annual/hourly reporting indicator retained from source schema.

11. OEWS/SOC Enrichment and Classification Fields

The finalized OEWS analytical layer adds classification metadata above the 52 preserved normalized fields. These fields identify whether an OEWS occupation code resolves to a formal SOC occupation or to a legitimate OEWS reporting category, identify the applicable SOC version, and expose occupation hierarchy codes/titles for drill-down. The layer also retains parallel technical metadata for OEWS-specific and historical reporting categories.

Classification	Rows in oews_annual_final
formal_soc	6,112,307
oews_hybrid_or_specific	47,063
oews_aggregate	16,237
historical_oews_hybrid	10,130
oews_specific_aggregation	6,699
oews_specific	3,955
Total	6,196,391

These categories are descriptive classifications, not exclusion rules. Valid OEWS rows are retained even when an occupation code is not a one-for-one formal detailed SOC code.

12. Historical OEWS/SOC Interpretation Notes

The OEWS series is structurally consistent enough to consolidate, but classification/reporting conventions change over time. AEDR validation found that using both SOC 2010 and SOC 2018 greatly reduces apparent code mismatches. Remaining nonmatches are concentrated in recognizable OEWS aggregate, hybrid, transition, and reporting categories rather than indicating corrupted observations.

Period	Share of OEWS rows not matching a formal detailed code in either SOC reference
2011	4.28%
2012–2016	1.57%–1.75%
2017–2018	2.29%–2.35%
2019–2020	4.01%–4.07%
2021–2025	0.92%–0.95%

Analytical rule: do not delete or recode these observations merely because they do not resolve to a formal detailed SOC code. Use the published OEWS code/title and the applicable SOC reference/crosswalk to understand its reporting context.

13. How to Use the Database

- Use qcew_annual_final for most human-readable QCEW analysis and visualization; use qcew_annual when the canonical normalized source structure is required.
- Filter QCEW by aggregation level before comparing observations. Rows at different geographic or industry resolutions are not directly interchangeable.
- Use NAICS vintage and hierarchy fields for historically appropriate industry drill-down.
- Use oews_annual_final for occupation-level analysis requiring the full OEWS/SOC classification metadata.
- Use oews_visualization_final for visualization and BI workflows; it preserves the same rows as oews_annual_final while presenting a curated 79-column layout.
- Use SOC references and the 2010→2018 crosswalk for historical interpretation, but do not use the one-to-many crosswalk as a direct analytical join unless duplication is explicitly handled.
- Treat QCEW and OEWS as complementary evidence. Similar-sounding employment or wage measures may have different populations, estimation methods, and grains.
- Do not assume exact QCEW/OEWS NAICS-code equality is a complete integration strategy; the programs have different reporting structures and grains.
- Preserve disclosure and flag fields in analysis. A suppressed or flagged value is not equivalent to zero.

14. Provenance and Engineering Principles

- Original BLS source files and imported source tables are preserved; normalization and enrichment are layered above them.
- Categorical identifiers are treated as identifiers, not quantities. Leading zeros and alphanumeric codes are preserved where required.
- Historical classification vintages are retained rather than forcing all observations into the newest NAICS or SOC system.
- Reference titles and hierarchy fields are descriptive enrichment; they do not replace original source codes.
- Analytical and visualization views preserve canonical row meaning unless a view explicitly documents filtering or aggregation.
- Audit and engineering findings are documented chronologically. Later discoveries amend earlier conclusions through new entries rather than silently rewriting prior entries.

15. Authoritative Sources

- U.S. Bureau of Labor Statistics — QCEW Overview
- U.S. Bureau of Labor Statistics — QCEW Handbook of Methods
- U.S. Bureau of Labor Statistics — QCEW Annual CSV File Layout
- U.S. Bureau of Labor Statistics — QCEW Classifications
- U.S. Bureau of Labor Statistics — QCEW Industry Classification / Historical NAICS vintages
- U.S. Bureau of Labor Statistics — QCEW NAICS Hierarchy Crosswalk
- U.S. Bureau of Labor Statistics — QCEW High-Level Industries
- U.S. Bureau of Labor Statistics — OEWS Program Overview
- U.S. Bureau of Labor Statistics — OEWS Handbook of Methods
- U.S. Bureau of Labor Statistics — Standard Occupational Classification (SOC)
- U.S. Bureau of Labor Statistics — 2010 and 2018 SOC structures, definitions, and crosswalk materials

16. AEDR Validation Status

QCEW + historical NAICS: validated and signed off for AEDR documentation/publication preparation.

OEWS + SOC: consolidated and validated. The canonical OEWS series and oews_annual_final each contain 6,196,391 rows; final view validation found zero missing occupation classifications and zero join-generated duplicate rows. The SOC reference layer includes validated 2010 and 2018 structures/definitions and official crosswalk compatibility testing.

oews_visualization_final: validated as a 79-column projection of oews_annual_final with no filtering, joins, aggregation, grouping, or deduplication. It is the preferred visualization-ready OEWS layer.

This document describes the completed detailed-labor database. The separate FRED labor/income time-series database should receive its own technical reference.